Notes for PzAP

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Invalid Date

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Preface

This is a Quarto book.

To learn more about Quarto books visit https://quarto.org/docs/books.

1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

2 PzAP bilješke, 14.10.2026.

Nastavak igre s `titanic` skupom podataka.

```
import pandas as pd
import seaborn as sns
```

```
df = sns.load_dataset("titanic")
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	dec
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	Na
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	Na
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	Na

2.1 Primjer

Prikazivanje samo nekih stupaca.

```
df[["who", "class", "age"]].head()
```

	who	class	age
0	man	Third	22.0
1	woman	First	38.0
2	woman	Third	26.0
3	woman	First	35.0
4	man	Third	35.0

2.2 Zadatak

Usporedite `class` (putnička klasa) i `deck` (paluba) stupce. Opišite u kakvom su odnosu.

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    category
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 80.7 KB
```

Unakrsna tablica (cross-tabulation) između stupaca 'class' i 'deck'

```
cross_tab = pd.crosstab(df['class'], df['deck'], margins=True)
display(cross_tab)
```

deck class	A	B	C	D	E	F	G	All
First	15	47	59	29	25	0	0	175
Second	0	0	0	4	4	8	0	16
Third	0	0	0	0	3	5	4	12
All	15	47	59	33	32	13	4	203

Ne bismo rekli da ovdje ima redundantnih podataka, ali dostupni podaci potvrđuju što je poznato o smještaju putnika na palubama Titanica:

> The Titanic was designed to enforce a rigid social hierarchy, with passengers segregated strictly by their ticket class.
>
> • First Class (The Upper Decks): First-class passengers occupied the most luxurious areas of the ship, primarily on A, B, and C Decks. They had exclusive access to the Boat Deck promenades, the gym, the squash court (on G Deck), and the Titanic Grand Staircase. Their cabins were closest to the lifeboats.
>
> • Second Class (Middle to Aft Decks): Second-class accommodations were located mainly toward the back (aft) of the ship, spanning D, E, and F Decks. They had their own dedicated smoking room, library, and dining saloon.
>
> • Third Class / Steerage (Lower Decks & Outer Ends): Third-class passengers were housed in the least desirable sections, subject to the noise and vibration of the propellers and engines. They were split by gender: single men slept in the bow (front) on D through G Decks, while single women and families were housed in the stern (back).

2.3 Zadatak

U nastavku nam neki stupci tablice neće biti potrebni pa ćemo ih obrisati.

```
cols = ["survived", "pclass", "embarked", "sibsp", "parch"]

for c in cols:
    try:
        del df[c]
    except:
        pass # ignore, already deleted.

df.head()
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone
0	male	22.0	7.2500	Third	man	True	NaN	Southampton	no	False
1	female	38.0	71.2833	First	woman	False	C	Cherbourg	yes	False
2	female	26.0	7.9250	Third	woman	False	NaN	Southampton	yes	True
3	female	35.0	53.1000	First	woman	False	C	Southampton	yes	False
4	male	35.0	8.0500	Third	man	True	NaN	Southampton	no	True

2.4 O stupcima i retcima

DataFrame objekti imaju dva indeksa: - `df.index` koji indeksira retke; - `df.columns` koji prebraja stupce

```
print(" ".join(df.columns))
```

```
sex age fare class who adult_male deck embark_town alive alone
```

U našem trenutnom primjeru retci su indeksirani brojevima od 0 do 890.

```
df.index
```

```
RangeIndex(start=0, stop=891, step=1)
```

Uskoro ćemo raditi s podacima kod kojih su mogući (ili zadani) nešto zanimljiviji indeksi. Za ovaj skup podataka nijedan stupac trenutno nije bolji kandidat za indeks.

Ako želimo pristupiti dijelu redaka preko njihovog indeksa, to možemo s `.iloc` metodom.

```
df.iloc[:5] # isti rezultat kao i .head()
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone
0	male	22.0	7.2500	Third	man	True	NaN	Southampton	no	False
1	female	38.0	71.2833	First	woman	False	C	Cherbourg	yes	False
2	female	26.0	7.9250	Third	woman	False	NaN	Southampton	yes	True
3	female	35.0	53.1000	First	woman	False	C	Southampton	yes	False
4	male	35.0	8.0500	Third	man	True	NaN	Southampton	no	True

```
df.iloc[10:20]
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone
10	female	4.0	16.7000	Third	child	False	G	Southampton	yes	False
11	female	58.0	26.5500	First	woman	False	C	Southampton	yes	True
12	male	20.0	8.0500	Third	man	True	NaN	Southampton	no	True
13	male	39.0	31.2750	Third	man	True	NaN	Southampton	no	False
14	female	14.0	7.8542	Third	child	False	NaN	Southampton	no	True

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone
15	female	55.0	16.0000	Second	woman	False	NaN	Southampton	yes	True
16	male	2.0	29.1250	Third	child	False	NaN	Queenstown	no	False
17	male	NaN	13.0000	Second	man	True	NaN	Southampton	yes	True
18	female	31.0	18.0000	Third	woman	False	NaN	Southampton	no	False
19	female	NaN	7.2250	Third	woman	False	NaN	Cherbourg	yes	True

2.5 Instalacija dodatnih biblioteka

```
try:
    import faker
except:
    print("faker module not installed.")
```

U Colab okruženju nije dostupna biblioteka `faker` koju ćemo koristiti za generiranje lažnih (sintetskih) podataka.

Možemo je instalirati pomoću `pip` alata. To možemo izvesti direktno iz bilježnice, ali prije naredbe ćemo dodati `!` kako bi se naredba izvršila u ljusci.

```
!pip install faker
```

```
Requirement already satisfied: faker in /home/marcelix/p3.14/lib/python3.14/site-
packages (40.43.0)
```

Sad možemo prikazati neke od mogućnosti ove biblioteke.

```
from faker import Faker
fake = Faker("hr_HR")
```

```
print(fake.address())
```

```
Kraljevec II. 9c/2
38557 Čakovec
```

2.6 Zadatak

Retci u tablici iz skupa podataka `titanic` predstavljaju putnike.
Dodajte stupac naziva `name` u `df` u kojem ćete putnicima dodijeliti izmišljena imena.

Pripazite na pridruživanje imena obzirom na spol (sex).

2.6.1 Rješenje.

Prvi pokušaj.

```
df["name"] = "BLANK"  #
df.head()
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone	name
0	male	22.0	7.2500	Third	man	True	NaN	Southampton	no	False	BLANK
1	female	38.0	71.2833	First	woman	False	C	Cherbourg	yes	False	BLANK
2	female	26.0	7.9250	Third	woman	False	NaN	Southampton	yes	True	BLANK
3	female	35.0	53.1000	First	woman	False	C	Southampton	yes	False	BLANK
4	male	35.0	8.0500	Third	man	True	NaN	Southampton	no	True	BLANK

```
df["name"]   = [ fake.name() for x in range(891) ]
df.head()
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone	name
0	male	22.0	7.2500	Third	man	True	NaN	Southampton	no	False	Jure Brlas
1	female	38.0	71.2833	First	woman	False	C	Cherbourg	yes	False	Ane Bezić
2	female	26.0	7.9250	Third	woman	False	NaN	Southampton	yes	True	Nikolina Cr
3	female	35.0	53.1000	First	woman	False	C	Southampton	yes	False	Mihael Fleg
4	male	35.0	8.0500	Third	man	True	NaN	Southampton	no	True	Lucija Topi

Drugi pokušaj.
(moramo uvažiti spol)

Uočimo da `faker` podržava generiranje muških i ženskih imena.

```
fake.name_male(), fake.name_female()
```

```
('Katica Blažek', 'Antonija Barišić')
```

```
def izmisli_ime(sex):
  """ Izmisli ime, ali u skladu sa spolom (sex). """
  if sex == "female":
    return fake.name_female()
  else:
    return fake.name_male()

izmisli_ime("female")
```

```
'Goran Blažević'
```

```
df["name"] = df.sex.apply(izmisli_ime)
df.head()
```

	sex	age	fare	class	who	adult_male	deck	embark_town	alive	alone	name
0	male	22.0	7.2500	Third	man	True	NaN	Southampton	no	False	Mate Hadži
1	female	38.0	71.2833	First	woman	False	C	Cherbourg	yes	False	Pero Vukov
2	female	26.0	7.9250	Third	woman	False	NaN	Southampton	yes	True	Dušan Zma
3	female	35.0	53.1000	First	woman	False	C	Southampton	yes	False	Milena Vral
4	male	35.0	8.0500	Third	man	True	NaN	Southampton	no	True	Slavica Kov

2.6.2 Komentar

Možete uočiti da `hr` implementacija imena prema spolu ne radi očekivano.

---

3 Zadaci za vježbu

3.1 Zadatak

- Provjerite ima li među putnicima maloljetne djece koja su putovala sama.
- Koja je starost najmlađe osobe koja je putovala sama?
- Nakon koje dobi se putnici više ne smatraju djecom (mlađi od 18, od 21 ili nešto treće)?

3.2 Zadatak

- Umjesto `name` stupca napravite dva posebna stupca `first_name` i `last_name` s odgovarajućim (generiranim) podacima.
- Preuredite `df` tako da `last_name` i `first_name` budu njezini prvi stupci.

3.3 Rješenja

TODO.

4 Summary

In summary, this book has no content whatsoever.

References

Knuth, Donald E. 1984. "Literate Programming." *Comput. J.* (USA) 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.